

42. $y'' - 2y' + 10y = g(x)$, $y(0) = 0$, $y'(0) = 0$, where

$$g(x) = \begin{cases} 20, & 0 \leq x \leq \pi \\ 0, & x > \pi \end{cases}$$

Discussion Problems

43. Consider the differential equation $ay'' + by' + cy = e^{kx}$, where a , b , c , and k are constants. The auxiliary equation of the associated homogeneous equation is

$$am^2 + bm + c = 0.$$

- (a) If k is not a root of the auxiliary equation, show that we can find a particular solution of the form $y_p = Ae^{kx}$, where $A = 1/(ak^2 + bk + c)$.
- (b) If k is a root of the auxiliary equation of multiplicity one, show that we can find a particular solution of the form $y_p = Axe^{kx}$, where $A = 1/(2ak + b)$. Explain how we know that $k \neq -b/(2a)$.
- (c) If k is a root of the auxiliary equation of multiplicity two, show that we can find a particular solution of the form $y = Ax^2e^{kx}$, where $A = 1/(2a)$.
44. Discuss how the method of this section can be used to find a particular solution of $y'' + y = \sin x \cos 2x$. Carry out your idea.
45. Without solving, match a solution curve of $y'' + y = f(x)$ shown in the figure with one of the following functions:
- | | |
|---------------------------|-------------------------|
| (i) $f(x) = 1$, | (ii) $f(x) = e^{-x}$, |
| (iii) $f(x) = e^x$, | (iv) $f(x) = \sin 2x$, |
| (v) $f(x) = e^x \sin x$, | (vi) $f(x) = \sin x$. |

Briefly discuss your reasoning.

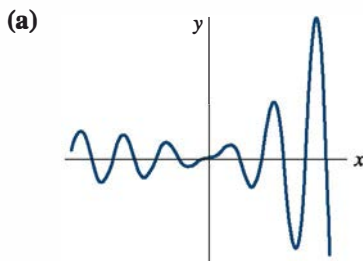


FIGURE 3.4.1 Solution curve

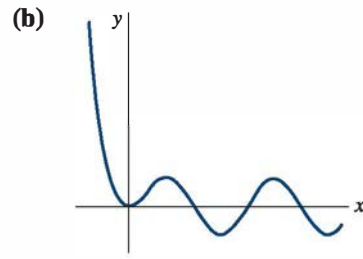


FIGURE 3.4.2 Solution curve

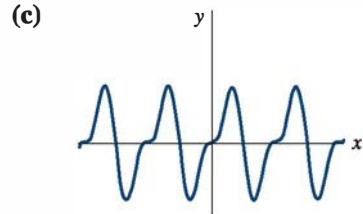


FIGURE 3.4.3 Solution curve

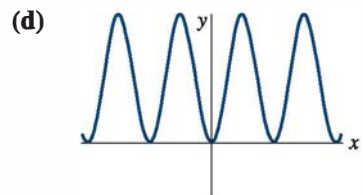


FIGURE 3.4.4 Solution curve

Computer Lab Assignments

In Problems 46 and 47, find a particular solution of the given differential equation. Use a CAS as an aid in carrying out differentiations, simplifications, and algebra.

46. $y'' - 4y' + 8y = (2x^2 - 3x)e^{2x} \cos 2x + (10x^2 - x - 1)e^{2x} \sin 2x$

47. $y^{(4)} + 2y'' + y = 2 \cos x - 3x \sin x$

3.5 Variation of Parameters

Introduction The method of variation of parameters used in Section 2.3 to find a particular solution of a linear first-order differential equation is applicable to linear higher-order equations as well. Variation of parameters has a distinct advantage over the method of the preceding section in that it *always* yields a particular solution y_p provided the associated homogeneous equation can be solved. In addition, the method presented in this section, unlike undetermined coefficients, is *not* limited to cases where the input function is a combination of the four types of functions listed on page 126, nor is it limited to differential equations with constant coefficients.

Some Assumptions To adapt the method of variation of parameters to a linear second-order differential equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x), \quad (1)$$

we begin as we did in Section 3.2—we put (1) in the **standard form**

$$y'' + P(x)y' + Q(x)y = f(x) \quad (2)$$

by dividing through by the lead coefficient $a_2(x)$. Equation (2) is the second-order analogue of the linear first-order equation $dy/dx + P(x)y = f(x)$. In (2) we shall assume $P(x)$, $Q(x)$, and $f(x)$ are continuous on some common interval I . As we have already seen in Section 3.3, there is no difficulty in obtaining the complementary function y_c of (2) when the coefficients are constants.

□ Method of Variation of Parameters Corresponding to the substitution $y_p = u_1(x)y_1(x) + u_2(x)y_2(x)$ that we used in Section 2.3 to find a particular solution y_p of $dy/dx + P(x)y = f(x)$, for the linear second-order DE (2) we seek a solution of the form

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x), \quad (3)$$

where y_1 and y_2 form a fundamental set of solutions on I of the associated homogeneous form of (1). Using the Product Rule to differentiate y_p twice, we get

$$\begin{aligned} y_p' &= u_1y_1' + y_1u_1' + u_2y_2' + y_2u_2' \\ y_p'' &= u_1y_1'' + y_1'u_1' + y_1u_1'' + u_2'y_2' + u_2y_2'' + y_2'u_2' + y_2u_2'' + u_2'y_2'. \end{aligned}$$

Substituting (3) and the foregoing derivatives into (2) and grouping terms yields

$$\begin{aligned} y_p'' + P(x)y_p' + Q(x)y_p &= u_1 \overbrace{[y_1'' + Py_1' + Qy_1]}^{\text{zero}} + u_2 \overbrace{[y_2'' + Py_2' + Qy_2]}^{\text{zero}} \\ &\quad + y_1u_1'' + u_1'y_1' + y_2u_2'' + u_2'y_2' + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' \\ &= \frac{d}{dx} [y_1u_1'] + \frac{d}{dx} [y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' \\ &= \frac{d}{dx} [y_1u_1' + y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' = f(x). \end{aligned} \quad (4)$$

Because we seek to determine two unknown functions u_1 and u_2 , reason dictates that we need two equations. We can obtain these equations by making the further assumption that the functions u_1 and u_2 satisfy $y_1u_1' + y_2u_2' = 0$. This assumption does not come out of the blue but is prompted by the first two terms in (4), since, if we demand that $y_1u_1' + y_2u_2' = 0$, then (4) reduces to $y_1'u_1' + y_2'u_2' = f(x)$. We now have our desired two equations, albeit two equations for determining the derivatives u_1' and u_2' . By Cramer's rule, the solution of the system

$$\begin{aligned} y_1u_1' + y_2u_2' &= 0 \\ y_1'u_1' + y_2'u_2' &= f(x) \end{aligned}$$

can be expressed in terms of determinants:

$$u_1' = \frac{W_1}{W} = -\frac{y_2f(x)}{W} \quad \text{and} \quad u_2' = \frac{W_2}{W} = \frac{y_1f(x)}{W} \quad (5)$$

where
$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}. \quad (6)$$

The functions u_1 and u_2 are found by integrating the results in (5). The determinant W is recognized as the Wronskian of y_1 and y_2 . By linear independence of y_1 and y_2 on I , we know that $W(y_1(x), y_2(x)) \neq 0$ for every x in the interval.

□ Summary of the Method Usually it is not a good idea to memorize formulas in lieu of understanding a procedure. However, the foregoing procedure is too long and complicated to use each time we wish to solve a differential equation. In this case it is more efficient to simply

◀ If you are unfamiliar with Cramer's rule, see Section 8.7.

use the formulas in (5). Thus to solve $a_2 y'' + a_1 y' + a_0 y = g(x)$, first find the complementary function $y_c = c_1 y_1 + c_2 y_2$ and then compute the Wronskian $W(y_1(x), y_2(x))$. By dividing by a_2 , we put the equation into the standard form $y'' + P y' + Q y = f(x)$ to determine $f(x)$. We find u_1 and u_2 by integrating $u_1' = W_1/W$ and $u_2' = W_2/W$, where W_1 and W_2 are defined as in (6). A particular solution is $y_p = u_1 y_1 + u_2 y_2$. The general solution of equation (1) is then $y = y_c + y_p$.

EXAMPLE 1 General Solution Using Variation of Parameters

Solve $y'' - 4y' + 4y = (x + 1)e^{2x}$.

SOLUTION From auxiliary equation $m^2 - 4m + 4 = (m - 2)^2 = 0$ we have $y_c = c_1 e^{2x} + c_2 x e^{2x}$. With the identifications $y_1 = e^{2x}$ and $y_2 = x e^{2x}$, we next compute the Wronskian:

$$W(e^{2x}, x e^{2x}) = \begin{vmatrix} e^{2x} & x e^{2x} \\ 2e^{2x} & 2x e^{2x} + e^{2x} \end{vmatrix} = e^{4x}.$$

Since the given differential equation is already in form (2) (that is, the coefficient of y'' is 1), we identify $f(x) = (x + 1)e^{2x}$. From (6) we obtain

$$W_1 = \begin{vmatrix} 0 & x e^{2x} \\ (x + 1)e^{2x} & 2x e^{2x} + e^{2x} \end{vmatrix} = -(x + 1)x e^{4x}, \quad W_2 = \begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & (x + 1)e^{2x} \end{vmatrix} = (x + 1)e^{4x},$$

and so from (5)

$$u_1' = -\frac{(x + 1)x e^{4x}}{e^{4x}} = -x^2 - x, \quad u_2' = \frac{(x + 1)e^{4x}}{e^{4x}} = x + 1.$$

It follows by integrating that $u_1 = -\frac{1}{3}x^3 - \frac{1}{2}x^2$ and $u_2 = \frac{1}{2}x^2 + x$. Hence

$$y_p = \left(-\frac{1}{3}x^3 - \frac{1}{2}x^2\right)e^{2x} + \left(\frac{1}{2}x^2 + x\right)x e^{2x} = \frac{1}{6}x^3 e^{2x} + \frac{1}{2}x^2 e^{2x}$$

$$\text{and} \quad y = y_c + y_p = c_1 e^{2x} + c_2 x e^{2x} + \frac{1}{6}x^3 e^{2x} + \frac{1}{2}x^2 e^{2x}. \quad \equiv$$

EXAMPLE 2 General Solution Using Variation of Parameters

Solve $4y'' + 36y = \csc 3x$.

SOLUTION We first put the equation in the standard form (2) by dividing by 4:

$$y'' + 9y = \frac{1}{4} \csc 3x.$$

Since the roots of the auxiliary equation $m^2 + 9 = 0$ are $m_1 = 3i$ and $m_2 = -3i$, the complementary function is $y_c = c_1 \cos 3x + c_2 \sin 3x$. Using $y_1 = \cos 3x$, $y_2 = \sin 3x$, and $f(x) = \frac{1}{4} \csc 3x$, we obtain

$$W(\cos 3x, \sin 3x) = \begin{vmatrix} \cos 3x & \sin 3x \\ -3 \sin 3x & 3 \cos 3x \end{vmatrix} = 3,$$

$$W_1 = \begin{vmatrix} 0 & \sin 3x \\ \frac{1}{4} \csc 3x & 3 \cos 3x \end{vmatrix} = -\frac{1}{4}, \quad W_2 = \begin{vmatrix} \cos 3x & 0 \\ -3 \sin 3x & \frac{1}{4} \csc 3x \end{vmatrix} = \frac{1}{4} \frac{\cos 3x}{\sin 3x}.$$

Integrating

$$u_1' = \frac{W_1}{W} = -\frac{1}{12} \quad \text{and} \quad u_2' = \frac{W_2}{W} = \frac{1}{12} \frac{\cos 3x}{\sin 3x}$$

gives $u_1 = -\frac{1}{12}x$ and $u_2 = \frac{1}{36} \ln |\sin 3x|$. Thus a particular solution is

$$y_p = -\frac{1}{12}x \cos 3x + \frac{1}{36}(\sin 3x) \ln |\sin 3x|.$$

The general solution of the equation is

$$y = y_c + y_p = c_1 \cos 3x + c_2 \sin 3x - \frac{1}{12} x \cos 3x + \frac{1}{36} (\sin 3x) \ln |\sin 3x|. \quad (7) \equiv$$

Equation (7) represents the general solution of the differential equation on, say, the interval $(0, \pi/3)$.

□ Constants of Integration When computing the indefinite integrals of u'_1 and u'_2 , we need not introduce any constants. This is because

$$\begin{aligned} y = y_c + y_p &= c_1 y_1 + c_2 y_2 + (u_1 + a_1) y_1 + (u_2 + b_1) y_2 \\ &= (c_1 + a_1) y_1 + (c_2 + b_1) y_2 + u_1 y_1 + u_2 y_2 \\ &= C_1 y_1 + C_2 y_2 + u_1 y_1 + u_2 y_2. \end{aligned}$$

EXAMPLE 3 General Solution Using Variation of Parameters

Solve $y'' - y = 1/x$.

SOLUTION The auxiliary equation $m^2 - 1 = 0$ yields $m_1 = -1$ and $m_2 = 1$. Therefore $y_c = c_1 e^x + c_2 e^{-x}$. Now $W(e^x, e^{-x}) = -2$ and

$$\begin{aligned} u'_1 &= -\frac{e^{-x}(1/x)}{-2}, & u_1 &= \frac{1}{2} \int_{x_0}^x \frac{e^{-t}}{t} dt, \\ u'_2 &= \frac{e^x(1/x)}{-2}, & u_2 &= -\frac{1}{2} \int_{x_0}^x \frac{e^t}{t} dt. \end{aligned}$$

Since the foregoing integrals are nonelementary, we are forced to write

$$y_p = \frac{1}{2} e^x \int_{x_0}^x \frac{e^{-t}}{t} dt - \frac{1}{2} e^{-x} \int_{x_0}^x \frac{e^t}{t} dt,$$

and so

$$y = y_c + y_p = c_1 e^x + c_2 e^{-x} + \frac{1}{2} e^x \int_{x_0}^x \frac{e^{-t}}{t} dt - \frac{1}{2} e^{-x} \int_{x_0}^x \frac{e^t}{t} dt. \quad \equiv$$

In Example 3 we can integrate on any interval $[x_0, x]$ not containing the origin. Also see Examples 2 and 3 in Section 3.10.

□ Higher-Order Equations The method we have just examined for nonhomogeneous second-order differential equations can be generalized to linear n th-order equations that have been put into the standard form

$$y^{(n)} + P_{n-1}(x)y^{(n-1)} + \cdots + P_1(x)y' + P_0(x)y = f(x). \quad (8)$$

If $y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$ is the complementary function for (8), then a particular solution is

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x) + \cdots + u_n(x)y_n(x),$$

where the u'_k , $k = 1, 2, \dots, n$, are determined by the n equations

$$\begin{aligned} y_1 u'_1 + y_2 u'_2 + \cdots + y_n u'_n &= 0 \\ y'_1 u'_1 + y'_2 u'_2 + \cdots + y'_n u'_n &= 0 \\ \vdots &\vdots \\ y_1^{(n-1)} u'_1 + y_2^{(n-1)} u'_2 + \cdots + y_n^{(n-1)} u'_n &= f(x). \end{aligned} \quad (9)$$

The first $n - 1$ equations in this system, like $y_1 u'_1 + y_2 u'_2 = 0$ in (4), are assumptions made to simplify the resulting equation after $y_p = u_1(x)y_1(x) + \cdots + u_n(x)y_n(x)$ is substituted in (8). In this case, Cramer's rule gives

$$u'_k = \frac{W_k}{W}, \quad k = 1, 2, \dots, n,$$

where W is the Wronskian of $y_1, y_2, \dots, y_n$ and W_k is the determinant obtained by replacing the k th column of the Wronskian by the column consisting of the right-hand side of (9), that is, the column $(0, 0, \dots, f(x))$. When $n = 2$ we get (5). When $n = 3$, the particular solution is $y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$, where y_1, y_2 , and y_3 constitute a linearly independent set of solutions of the associated homogeneous DE, and u_1, u_2, u_3 are determined from

$$u'_1 = \frac{W_1}{W}, \quad u'_2 = \frac{W_2}{W}, \quad u'_3 = \frac{W_3}{W}, \quad (10)$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y'_2 & y'_3 \\ f(x) & y''_2 & y''_3 \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y'_1 & 0 & y'_3 \\ y''_1 & f(x) & y''_3 \end{vmatrix}, \quad W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y'_1 & y'_2 & 0 \\ y''_1 & y''_2 & f(x) \end{vmatrix}, \quad \text{and } W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y'_1 & y'_2 & y'_3 \\ y''_1 & y''_2 & y''_3 \end{vmatrix}.$$

See Problems 25 and 26 in Exercises 3.5.

Remarks

In the problems that follow do not hesitate to simplify the form of y_p . Depending on how the antiderivatives of u'_1 and u'_2 are found, you may not obtain the same y_p as given in the answer section. For example, in Problem 3 in Exercises 3.5, both $y_p = \frac{1}{2} \sin x - \frac{1}{2}x \cos x$ and $y_p = \frac{1}{4} \sin x - \frac{1}{2}x \cos x$ are valid answers. In either case the general solution $y = y_c + y_p$ simplifies to $y = c_1 \cos x + c_2 \sin x - \frac{1}{2}x \cos x$. Why?

3.5 Exercises

Answers to selected odd-numbered problems begin on page ANS-5.

In Problems 1–18, solve each differential equation by variation of parameters.

1. $y'' + y = \sec x$
2. $y'' + y = \tan x$
3. $y'' + y = \sin x$
4. $y'' + y = \sec \theta \tan \theta$
5. $y'' + y = \cos^2 x$
6. $y'' + y = \sec^2 x$
7. $y'' - y = \cosh x$
8. $y'' - y = \sinh 2x$
9. $y'' - 4y = \frac{e^{2x}}{x}$
10. $y'' - 9y = \frac{9x}{e^{3x}}$
11. $y'' + 3y' + 2y = \frac{1}{1 + e^x}$
12. $y'' - 2y' + y = \frac{e^x}{1 + x^2}$
13. $y'' + 3y' + 2y = \sin e^x$
14. $y'' - 2y' + y = e^t \arctan t$
15. $y'' + 2y' + y = e^{-t} \ln t$
16. $2y'' + 2y' + y = 4\sqrt{x}$
17. $3y'' - 6y' + 6y = e^x \sec x$
18. $4y'' - 4y' + y = e^{x/2} \sqrt{1 - x^2}$

In Problems 19–22, solve each differential equation by variation of parameters subject to the initial conditions $y(0) = 1, y'(0) = 0$.

19. $4y'' - y = xe^{x/2}$
20. $2y'' + y' - y = x + 1$
21. $y'' + 2y' - 8y = 2e^{-2x} - e^{-x}$
22. $y'' - 4y' + 4y = (12x^2 - 6x)e^{2x}$

In Problems 23 and 24, the indicated functions are known linearly independent solutions of the associated homogeneous differential equation on the interval $(0, \infty)$. Find the general solution of the given nonhomogeneous equation.

23. $x^2 y'' + xy' + (x^2 - \frac{1}{4})y = x^{3/2}$; $y_1 = x^{-1/2} \cos x$, $y_2 = x^{-1/2} \sin x$
24. $x^2 y'' + xy' + y = \sec(\ln x)$; $y_1 = \cos(\ln x), y_2 = \sin(\ln x)$